

Case Study 2 Group 4

Bryanda Meza Amillano, Olga Carrillo, Lamek Chiwaka, Sadiskshya Dahal

2023-02-08

Introduction

Our report aims to identify potential indicators that can reduce the murder rate in [US State] based on a comprehensive analysis of crime.rdata from all 50 US states. The crime.rdata data set includes the following variables by all 50 US states: total state population, the median age in the state, percentage of unemployed labor force, firearm legislation score (higher scores indicate more legislation), robbery, burglary, aggravated assault, larceny/theft and motor vehicle theft rates per 100,000 inhabitants, and number of murders including non negligent manslaughter. Given the complexity of neighborhood violence, an ecological approach is essential for understanding and reducing the murder rate within the state. Building on Hureau's (2022) recommendation, this study aims to investigate the relationship between US murder rates and various crime-related factors, including age and unemployment. Specifically, we seek to answer the following questions: 1) How do age and unemployment affect crime rates? 2) Which combinations of variables have the most significant impact on the US murder rate? The associations and relationships shown will help determine what initiatives are worth funding. Given our limited budget, we will prioritize initiatives that are most likely to reduce the murder rate within the state, taking into account their feasibility and cost-effectiveness.

Methods

Data analysis was conducted using R programming language version 2022.07.01. To begin the analysis, we downloaded and installed the necessary libraries for this project. These libraries were required to run specific functions for finding the best-fit model, plotting the variables within the model, and creating a heat map for crude murder rates by state. The crime data set was loaded, and initial data exploration was performed to obtain summary statistics for each variable, total number of observations and variables, and the first six rows within the data frame, respectively. Preliminary histograms of each variable were also created for early data exploration. To enable fair comparison, the murder rate was multiplied by 100,000 inhabitants to create a new variable labeled "Murder". A color-coded heat map of the United States was generated using the crime data set to show crude murder rates by state. The best linear model was selected using the regsubsets function in R, followed by the use of scatter plots generated by ggplot to visualize the relationships between the murder rate and the other variables in the crime data set. Assumptions, including linearity, independence, normal distribution, and equal variances, were checked for both uni variate and multivariate models. Linear models 2 and 6 were uni variate models, and linear model 4 was multivariate. Interaction was checked for the multivariate models as well. To account for the limited budget, we utilized statistical methods to gain a deeper understanding of the selected variables, including line of best fit, r coefficient, and Pearson Correlation R code. The best model fit was determined by comparing adjusted R-squared and AIC.

Results

The crime data set included 50 observations and 12 variables. The heat map identified that Missouri, Louisiana, Mississippi, and South Carolina had higher murder rates than other states. After an exhaustive search for the highest adjusted R-squared, the best model fit for the murder rate model included variables percentage of unemployed labor force (Unemployment), the median age in the state (Age), firearm legislation score-higher scores indicate more legislation (BradyScore), Robbery, Burglary, Aggravated Assault, and Motor Vehicle Theft rates per 100,000 inhabitants. The adjusted R-squared for this model, which was the

highest, was 0.6457 with an AIC of 176.968. When running assumptions, it was assumed independence was met. In linear model 1, which was the chosen model, linearity was met for all seven variables, normality showed multiple peaks and slight right-skew, and the equal variance plots were relatively spread out with some influential outliers. Motor vehicle theft, robbery, and assault all showed to be significant with p-values of 0.01, 0.003, and 0.02. Controlling for unemployment, BradyScore, age, robbery, assault, and burglary, the difference in murder rate comparing a motor vehicle theft rate that differs by 1 is -.007 per 100,000. Controlling for unemployment, BradyScore, age, assault, and burglary, motor vehicle theft the difference in murder rate comparing a robbery rate that differs by 1 is .02 per 100,000. Controlling for unemployment, BradyScore, age, robbery, and burglary, motor vehicle theft the difference in murder rate comparing an assault rate that differs by 1 is .006 per 100,000. Linear model 2 was a uni variate model and scatter plot showed a linear moderate positive association between murder rate and assault rate, normality showed multiple peaks but had a relatively normal shape, and equal variances plot was not equally spread for murder, but assault was evenly spread. Assault in this model was also significant with a p-value less than 0.05. Linear model 6 was also univariate and linearity showed a moderate linear positive association between murder rate and robbery rate, normality showed multiple peaks with a relatively normal shape, and equal variances were evenly spread out for robbery but had some influential outliers. Robbery was also significant in this model with a p-value less than 0.05. Linear model 4 was the multivariate model. Linearity was met for Murder~Unemployment*Robbery. Normality for model 4 showed multiple peaks with a relatively normal shape. Equal variances for unemployment and assault were met. Linear model 4 appeared to show interaction between murder and robbery with unemployment.

Discussion and Conclusion

The case study approach ultimately resulted in three multivariable linear regression models of interaction based on preliminary R^2 and AIC analyses for all the variables. Whether a univariable or multivariable model is best is frivolous within the context of this case study as it is critical to recognize the relationship between the murder rate and each dependent variable and ultimately the effect of combining the variables on the murder rate. If the focus is solely on unemployment exposure, it is significant to understand which factors interact with unemployment, in this respect, assault, robbery, and car theft.

The results based on linear regression model 4 suggest that the relationship between certain types of crime and murder is moderated by unemployment rates. According to linear regression model 4, the association between robbery and murder is different by unemployment. This may suggest that areas with higher rates of unemployment and robbery may be at greater risk for higher murder rates.

This outcome concurs with earlier studies that have discovered a connection between unemployment and crime rates. For example, Raphael and Winter-Ebmer (2001) found that unemployment has a positive correlation with crime rates. Additionally, Nunley et al., demonstrated that unemployment has a significant impact on the murder rate, specifically that age (15-29) and unemployment (with inflation) indicate a positive relationship with murder rates in the US (2011).

The relationship between unemployment and crime rates may be more complex, as demonstrated by per, Kearney et al., that socioeconomically disadvantaged youth are swayed more towards crime because of low employment opportunities and perceived threats due to increased risk of abuse (2014). Further research is required to better comprehend the relationship between unemployment and crime rates, as well as the various factors that may influence crime. The data on interaction can provide probable hypotheses why unemployment, assault, robbery, and theft are related and potential solutions. For example, would the implementation of unemployment safety nets that offer educational resources on how to avoid committing crimes for monetary gain, along with an explanation of the benefits, be effective in discouraging engagement in criminal activities, including murder?

A strength of this case study was that the dataset did not have missing data so associations between variables could more easily be observed in the results of the analysis. Another strength of the case study included a detailed description of the assumptions that needed to be verified for the linear models to ensure validity. Additionally, it comprehensively examined the adjusted r-squared findings to identify the best model fit per AIC.

However, our study had some limitations that should be considered when interpreting the results. Firstly, we only included variables that could potentially be associated with murder, which may have excluded some important factors. Secondly, due to the size of our dataset, we were unable to include all possible explanatory variables. Additionally, we had to convert the murder count to a rate in order to make it proportional for all states, which may have introduced some measurement error. A couple of the associated weaknesses with the case study is it did not analyze a specific state but rather a national rate. Due to significant socio-economic differences not only between states but also regionally, it would be best to analyze factors based on a specific state. In addition, the case study did not consider confounders, therefore subsequent steps include US murder linear regression models per state, as well as the analysis of potential confounders.

Unemployment is complex and can be associated with stress, decreased life satisfaction, and problems with alcohol abuse. Unemployment is also related to increased criminal activities, and interventions to reduce the number of unemployed can lead to a low crime rate in the state. We recommend that resources be set aside to improve the mental health of unemployed people. People will be taught how to deal with the stress of unemployment and not to dwell on the negative thoughts of unemployment (ABCT, n.d). We also recommend that the state streamline the unemployment application process and increase the bi-weekly payments. Policies that favor small businesses should be implemented as this has been shown to reduce unemployment (Dilanchiev, 2014).

Tables and Figures

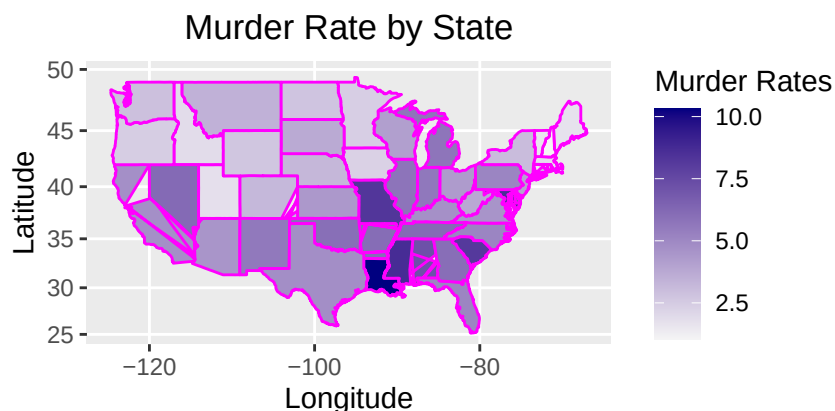


Table 1: Linear Model 2 (Murder and Assault)

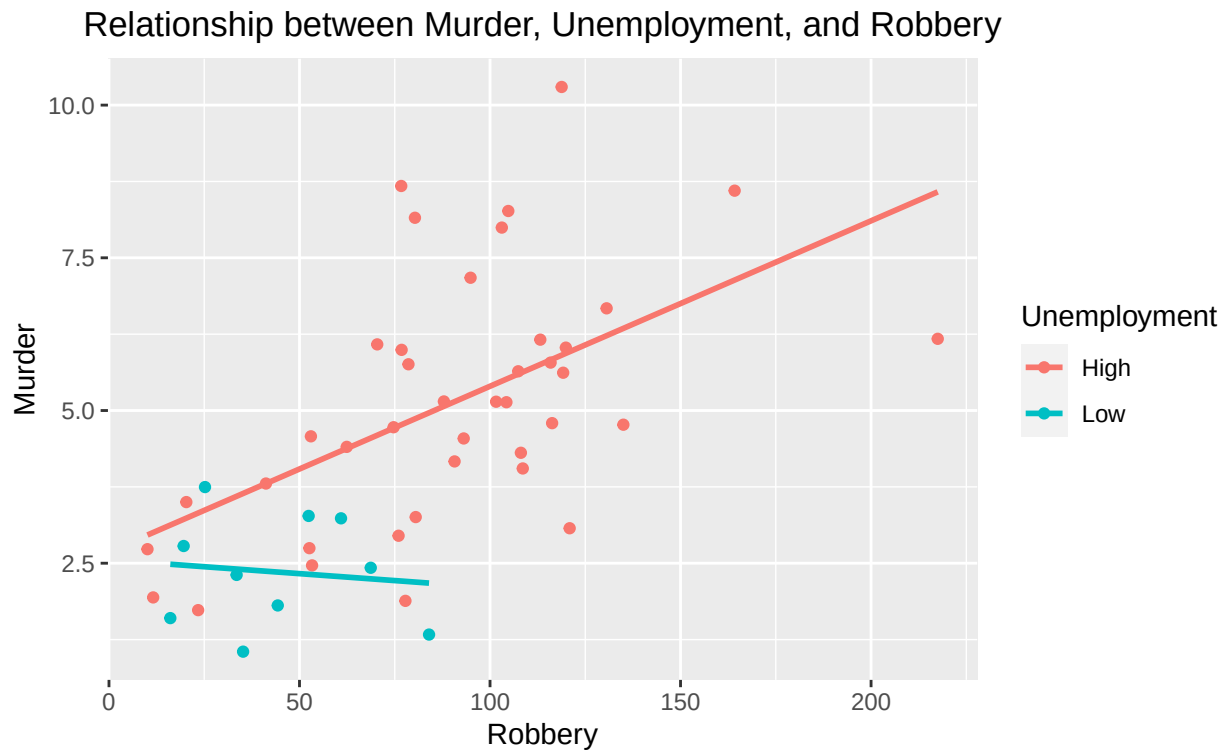
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.37	0.6	2.28	0.03	0.16	2.57
Assault	0.01	0.0	5.84	0.00	0.01	0.02

Table 2: Linear Model 6 (Murder and Robbery)

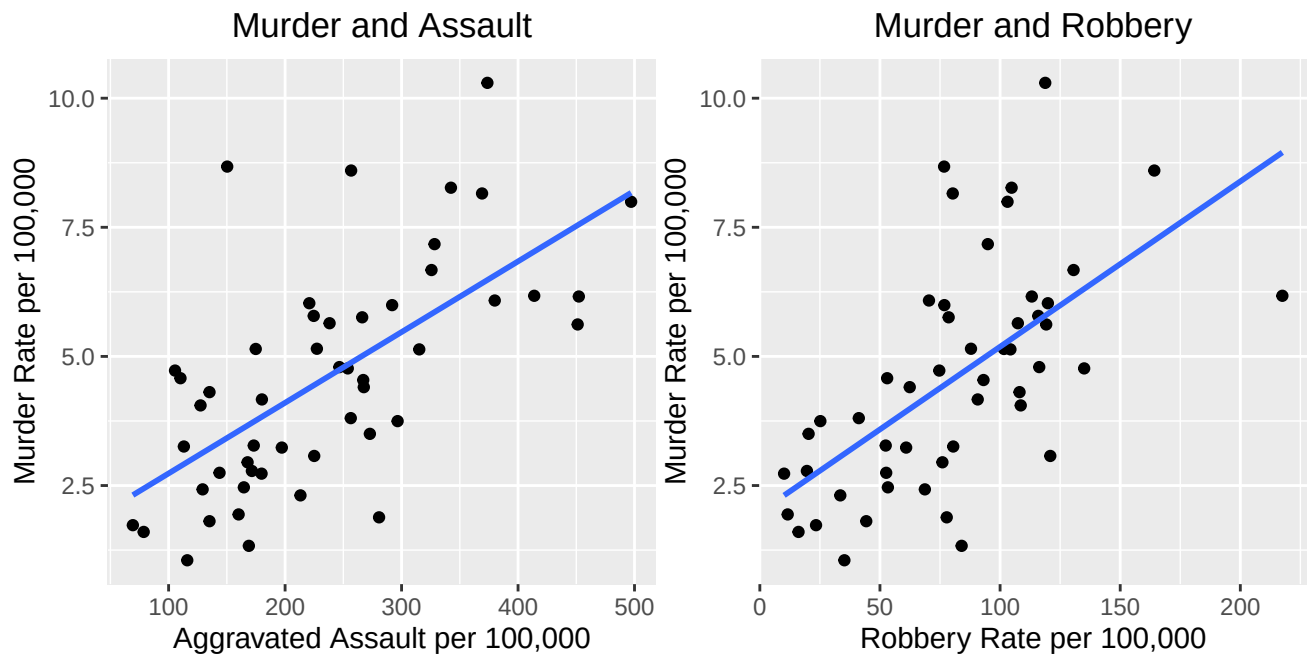
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.99	0.54	3.67	0	0.90	3.07
Robbery	0.03	0.01	5.36	0	0.02	0.04

Table 3: Linear Model 4 (Murder, Unemployment, Robbery)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	2.69	0.64	4.19	0.00	1.40	3.98
UnemploymentLow	-0.13	1.36	-0.10	0.92	-2.86	2.60
Robbery	0.03	0.01	4.15	0.00	0.01	0.04
UnemploymentLow:Robbery	-0.03	0.03	-1.25	0.22	-0.08	0.02



Appendix



References

- Association for behavioral and cognitive therapies (n.d). Unemployment. <https://www.abct.org/fact-sheets/unemployment/>.
- Dilanchiev, A. (2014). Relationship between Entrepreneurship and Unemployment: The Case of Georgia. *Journal of Social Sciences*, 3, 5-9.
- Hureau, D. M. (2022). Seeing Guns to See Urban Violence: Racial Inequality & Neighborhood Context. *Daedalus*, 151(1), 49-66. <https://www.jstor.org/stable/48638129>
- Kearney, M. S., Harris, B. H., Jácome, E., & Parker, L. (2014, May). Ten Economic Facts about Crime and Incarceration in the United States . Retrieved from https://www.brookings.edu/wp-content/uploads/2016/06/v8_THP_10CrimeFacts.pdf
- Nunley, J. M., Seals Jr, R. A., & Zietz, J. (2011). Demographic change, macroeconomic conditions, and the murder rate: The case of the United States, 1934-2006. *The Journal of Socio-Economics*, 40(6), 942-948.
- Raphael, S., & Winter-Ebmer, R. (2001). Identifying the Effect of Unemployment on Crime. *The Journal of Law & Economics*, 44(1), 259-283. <https://doi.org/10.1086/320275>
- TED. (2016). Inside the mind of a master procrastinator [Video file]. <https://www.youtube.com/watch?v=IScjygOnO0w>
- Tonry, M. (2014). Why Crime Rates Are Falling throughout the Western World. *Crime and Justice*, 43(1), 1-63. <https://doi.org/10.1086/678181>
- <https://www.youtube.com/watch?v=IScjygOnO0w>
- code from: <https://rpubs.com/knm6/mapsI> -> for heat map